



The influence of scene tilt on saccade directions is amplitude dependent

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ABSTRACT

When exploring a visual scene, humans make more saccades in the horizontal direction than any other direction. While many have shown that the horizontal saccade bias rotates in response to scene tilt, it is unclear whether this effect depends on saccade amplitude. We addressed this question by examining the effect of image tilt on the saccade direction distributions recorded during freely viewing natural scenes. Participants ($n = 20$) viewed scenes tilted at -30° , 0° , and 30° . Saccade distributions during free viewing rotated by an angle of $12.1^\circ \pm 6.7^\circ$ ($t(19) = 8.04$, $p < 0.001$) in the direction of the image tilt. When we partitioned the saccades according to their amplitude we found that small amplitude saccades occurred most in the horizontal direction while large amplitude saccades were more oriented to the scene tilt ($p < 0.001$). To further study the characteristics of small saccades and how they are affected by scene tilt, we looked at the effect of image tilt on small fixational saccades made while fixating a central target amidst a larger scene and found that fixational saccade distributions did not rotate with scene tilt ($-0.3^\circ \pm 1.7^\circ$ degrees; $t(19) = -0.8$, $p = 0.39$). These results suggest a combined effect of two reference frames in saccade generation: one egocentric reference frame that dominates for small saccades, biases them horizontally, and may be common for different tasks, and another allocentric reference frame that biases larger saccades along the orientation of an image during free viewing.

1. Introduction

When humans explore a visual scene, they use saccades – quick, ballistic eye movements – to fixate objects of interest and bring certain aspects of the scene onto the fovea. While saccades occur in all directions, they occur more often in the cardinal directions than the oblique direction and more in the horizontal than vertical direction. This pattern has been noted in a variety of tasks including free viewing [1,2,9,33], visual search [11,32], movie watching [5], and fixation [33].

The existence of the saccade direction bias is in part thought to be attributed to a combination of low-level (salient) and high-level (semantic) features of natural scenes. Decades of work on the statistics of natural scenes has shown that there is a tight coupling between scene structure and eye guidance as a result of contrast, edges, spatial frequency distributions, and orientation energy [23–25,41]. Other studies have elaborated this work to show that gaze is significantly impacted by relevant task goals, gist, and pertinent semantic content [17,19,50]. Because of the relationship between scene content and eye movements, the orientation of a scene changes the way we look at it. Work by Foulsham et al. [9] has shown that scene tilt affects saccade directions

and rotates the corresponding saccade direction distribution bias to align with the orientation of the scene. This result has been replicated using a variety of frames with tilted images [1] and using tilted images that take up the entire field of view [2]. Others have even found that saccade directions are influenced by oriented noisy textures and gratings [52]. This body of work provides support for the idea that the saccade direction bias may in part be generated with respect to the orientation of the scene or image itself.

Previous work has found that regions of natural scenes that are fixated have different statistics than regions of scenes that are not fixated [38,45], and even more interestingly, that the regions of scenes following long saccades have different statistics than the regions of scenes following short saccades [46]. Specifically, the difference between fixated and non-fixated edge and contrast information decreases as a function of saccade size, indicating that high spatial frequency information is important for short saccades and less so for long saccades. Some have speculated that short and long saccades are generated by different brain mechanisms [10], though more recent studies have shown evidence that a common saccade generator is responsible for both small saccades made during fixation and large saccades [14,33,40].

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Others have shown that large saccades are more useful than small saccades at decorrelating neural input since regions of scenes that are far away from each other are more likely to have different properties [43]. To our knowledge, no one yet has examined whether the influence of a tilted scene on saccade directions changes with saccade amplitude.

Directional biases of small saccades made during fixation have been shown previously [6,22,33,51]. There are several factors that may influence the direction of fixational saccades, such as peripheral cues and the type of fixation spot. Previous studies have found that attention to certain peripheral areas during fixation prompts the occurrence of fixational saccades in the direction of or away from the attended location [7,13,20,39], though see [21,51] for rebuttal. Others have studied whether the type of fixation spot influences fixation stability and have found that certain fixation spots promote tighter distributions [3,48]. Visual vestibular cues for gravity, such as a tilted image, might influence the direction of small fixational saccades. It is known that a tilted image induces altered perceptions of upright [15,16] in addition to invoking optostatic torsion [36]. A recent study examined microsaccade directions made during fixation in response to oriented gratings, and interestingly found no effect of the tilted gratings [49]. It could be that gratings simply do not contain sufficient salient or semantic content to warrant a reorientation of the saccade direction bias.

Here, we use tilted natural scenes to examine whether saccade amplitude plays a role in determining saccade directions. We use different tasks, free viewing, and fixation, to experimentally manipulate saccade amplitude and then test whether saccade direction distributions of saccades of different sizes reorient in response to scene tilt.

2. Methods

2.1. Participants

Twenty individuals (ages 21–40 with an average of 25 years; 17 female, 3 male) from the University of California Berkeley participated in the study. Written informed consent was obtained from each individual prior to study participation. The research followed the tenets of the Declaration of Helsinki and the Institutional Review Board of the university approved the study.

Prior to conducting the full study, we implemented an a priori one-tailed *t*-test power analysis using G*Power [8] on pilot data, which revealed that at least 15 participants were required for the study based on an effect size of 0.9 (supported by data showing an average of 8.3° of effect and 9.1° SD during free viewing of a tilted scene), an alpha of 0.05, and a power of 0.8.

2.2. Procedure

Participants were seated 90 cm from an 4 K OLED screen (LG LED65CSPUA 3840 × 2160 pixels, 144.4 × 82.2 cm, 60 Hz refresh rate). Participants rested their heads on a chin-rest and were eye tracked using a custom-built binocular eye tracker running at 250 Hz [34,42]. The experimental paradigm was run using custom scripts in MATLAB and Psychtoolbox [4,37].

Each trial was 12 s in duration and contained a period of fixation (two seconds) followed by a period of the task (10 s, either fixation or free viewing). Participants knew to either fixate or free view on each trial by the presence or absence of a fixation dot overlaid on top of a scene. The fixation dot (present on fixation trials) was 0.5° and the scene (present on all trials regardless of task) was 20° × 20°. Natural scenes could be tilted either −30°, 0°, or 30° and were obtained from the MIT CSAIL database [26]. The experiment was fully-crossed and randomized, which resulted in 240 trials (2 tasks × 3 scene tilts × 40 scenes) per participant. The experiment lasted about 1.5 h.

Each subject participated in a debriefing interview following the study that consisted of five questions regarding the subject's experience in completing the study. One question was: "What did you think we were

testing in the experiment." Seven out of 20 individuals were able to approximately guess our research question: whether the orientation of the scene influences eye movements.

2.3. Data analyses and statistics

Binocular eye position data was processed with custom scripts in MATLAB. Saccades were detected post-processing using an algorithm designed by Engbert and Kliegl [7] that is based on the robust standard deviation of eye movement data (with parameter $\lambda = 8$). We confirmed accurate saccade detection with visualizations of individual subjects' eye position traces (Fig. 1C and F) in addition to summary plots of the main sequence (Fig. 1A and D) and amplitude distributions (Fig. 1B and E).

We calculated polar distributions of saccade directions by applying a circular kernel density estimate (KDE) to the data [31]. The kernel was a wrapped Gaussian 0.1 rad in width that was applied from 0° to 360° in steps of 0.1°. We cross-correlated pairs of polar distributions for individual subjects (e.g., a polar histogram with scene tilt and one upright, see Results) to find the angular displacement in degrees (Δ) that resulted in the maximum correlation. The cross-correlation procedure was done circularly, to avoid edge effects, by repeating the KDE three times and searching within a range from −360° to 720°. These direction distribution displacements for individual subjects were bootstrapped to obtain 95% confidence intervals.

The reference frame index was calculated according to the equation $RIndex = \frac{\Delta_{30} + (-\Delta_{-30})}{30}$. Briefly, we reversed the values corresponding to the leftward image tilt (−30°), took the mean direction distribution displacement for left and right tilt, and then scaled the values by 30 (the absolute scene tilt amount in degrees) for each subject. Thus, the reference frame index could range from 0 to 1 where 0 indicates an ego-centered reference frame and 1 indicates a scene-centered or image-centered reference frame. The RIndex allows us to combine the data from left and right tilts and summarize it into a single number to test the overall significance of our effects.

We used *t*-tests and a linear mixed effects model to determine whether there were differences among select experimental conditions (see Results). We chose an alpha of 0.05 to determine significance.

3. Results

While previous work has shown that the saccade direction bias rotates in response to natural scene tilt, it is unknown whether this effect varies for saccades of different amplitudes. To answer this research question, we analyzed saccades during free viewing to determine if the direction of small and large saccades are influenced equally by the tilt of the scene. We then analyzed fixational saccade data to determine if saccades made during fixation were influenced by background scene tilt.

3.1. Saccade direction distributions during free viewing

With the aim of confirming previous work, we first asked whether saccade direction distributions during free viewing would be impacted by scene tilt. For this analysis, we focused on all saccades that were made during free viewing of a natural scene that was tilted −30°, 0°, or 30°. We calculated polar distributions of saccade directions and implemented a circular cross-correlation technique to determine the angular offset that resulted in the maximum correlation between pairs of distributions (see Methods). Fig. 2 shows that many subjects made saccades in alignment with the orientation of the scene, consistent with previous work (Fig. 2A,B). The average saccade direction distribution displacement across all subjects was 11.4° ($t(19) = 8.73$, $p < 0.001$) when comparing 0° and 30° scene tilts and −12.9° ($t(19) = -8.88$, $p < 0.001$) when comparing 0° and −30° scene tilts (Fig. 2C). The average reference frame index was 0.40, which was significantly different from zero

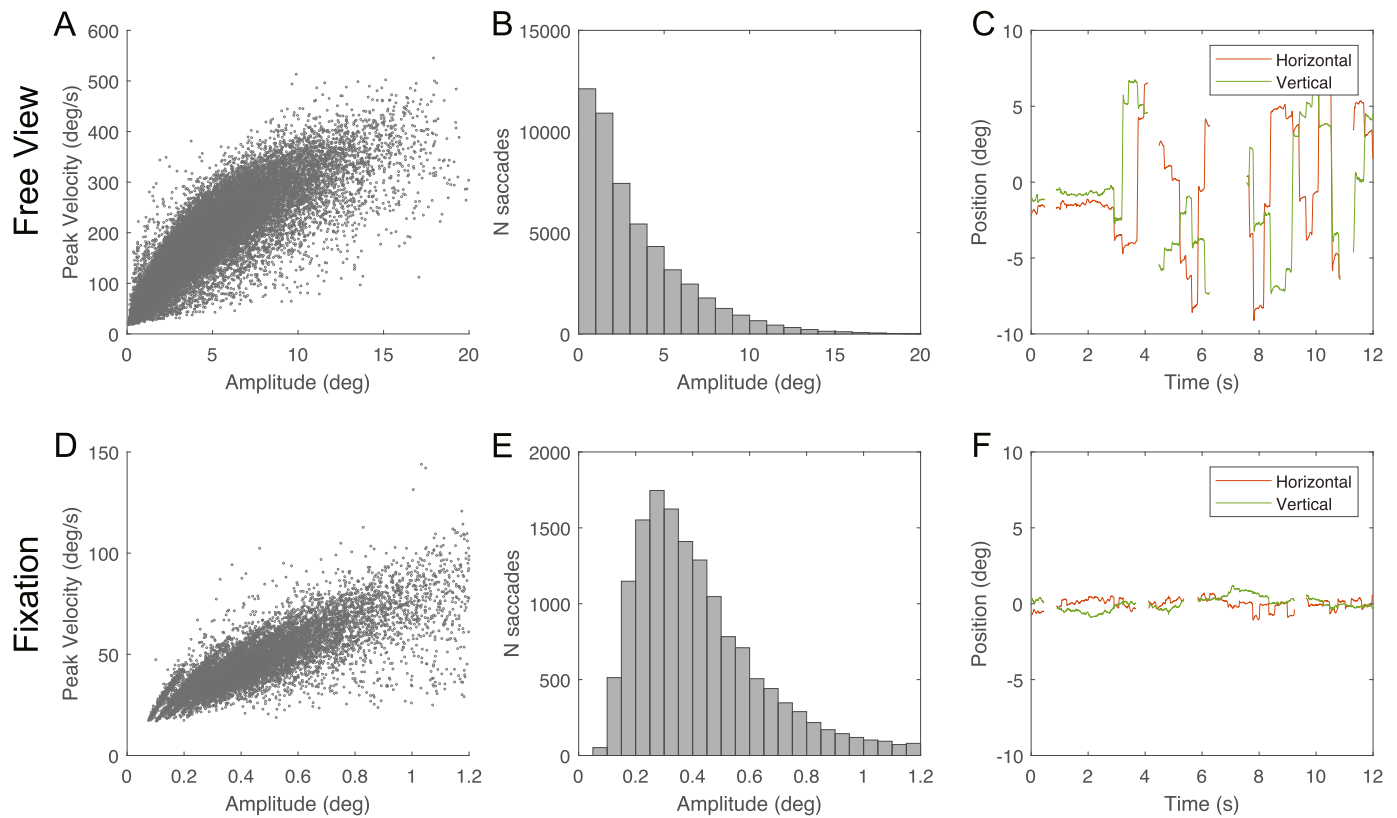


Fig. 1. Saccade metrics and eye movement traces for all eye movement data. Top row shows saccade data during free viewing while bottom row shows fixational saccade data during fixation. A and D) Main sequence. B and E) Histogram of saccade and fixational saccade amplitudes. C and F) Raw eye movement traces from one trial of one subject.

(indicating alignment with an egocentric reference frame; $t(19) = 8.04$, $p < 0.001$) and one (indicating alignment with an image reference frame; $t(19) = 11.84$, $p < 0.001$). This suggests that saccade directions were not exactly oriented with respect to the tilted scene or the upright world, head, or body.

3.2. Saccades direction biases are amplitude dependent during free viewing

While previous work has shown that saccade directions during free viewing occur along the primary axis of a tilted scene, it is unclear whether this is true across all saccade sizes. To determine whether the effect of image tilt on saccade directions during free viewing is amplitude-dependent, we partitioned the free viewing saccade data into four groups according to saccade amplitude. Thresholds for this partitioning were set based on quartiles to ensure that the same number of saccades were present in each group. Fig. 3 shows saccade direction distributions as a function of amplitude (Fig. 4A–4D) with the associated reference frame index for each amplitude quartile (Fig. 4E). The smallest quartile of free viewing saccades – those smaller than 1.1° – were primarily horizontal in direction and had an average reference frame index of 0.07 (95% CI = [0.003, 0.15]). The second quartile of saccades had a reference frame index of 0.20 (95% CI = [0.13, 0.27]), the third quartile had a reference frame index of 0.45 (95% CI = [0.38, 0.52]), and the largest quartile of saccades – those bigger than 4.7° – had an average reference frame index of 0.56 (95% CI = [0.49, 0.63]). We fitted a linear mixed effects model to the relationship between the amplitude of the saccade quartiles and the reference frame index including random slopes and intercepts. This model revealed that the reference frame indices were significantly related to the saccade amplitudes (Coeff = 0.07, 95% CI [0.04, 0.09], $p < 0.001$). These results suggest that larger saccades during free viewing were in closer alignment with the orientation of the

scene while smaller saccades during free viewing were in closer alignment with the upright head.

3.3. Saccade direction distributions during fixation

In the previous results, we observed that the directions of small saccades during free viewing occur primarily in the horizontal direction and are not affected by scene tilt. We wondered whether small saccades during a different task would be horizontally biased and unaffected by the tilt of the scene. Thus, we examined the small fixational saccades that occur during fixation of a central, round dot while a natural scene was displayed in the background/periphery that was tilted -30° , 0° , or 30° . We calculated polar distributions of saccade directions and implemented a circular cross-correlation technique to determine the angular offset that resulted in the maximum correlation between pairs of distributions (see Methods). As shown previously, we found that the background images had a small effect in the torsional position of the eyes by rotating them in the direction of the scene (average optostatic torsion = 0.22° , $p < 0.001$) [36]. Optostatic torsion occurred even though subjects primarily pay attention to the fixation spot to maintain fixation, suggesting that the background images were not completely ignored by the visual and ocular motor systems. Fig. 4 shows that subjects made fixational saccades that were largely horizontal in direction (Fig. 4A and B). The average fixational saccade direction distribution displacement across all subjects was 0.4° ($t(19) = 0.45$, $p = 0.65$) when comparing 0° and 30° scene tilts and 1.1° ($t(19) = 1.62$, $p = 0.12$) when comparing 0° and -30° scene tilts (Fig. 4C). The average reference frame index was -0.01 , which was not significantly different from zero (indicating an egocentric reference frame; $t(19) = -0.87$, $p = 0.39$) but was significantly different from one (indicating an image-centered reference frame; $t(19) = 77.69$, $p < 0.001$). This suggests that fixational saccades were made definitively with respect to the upright

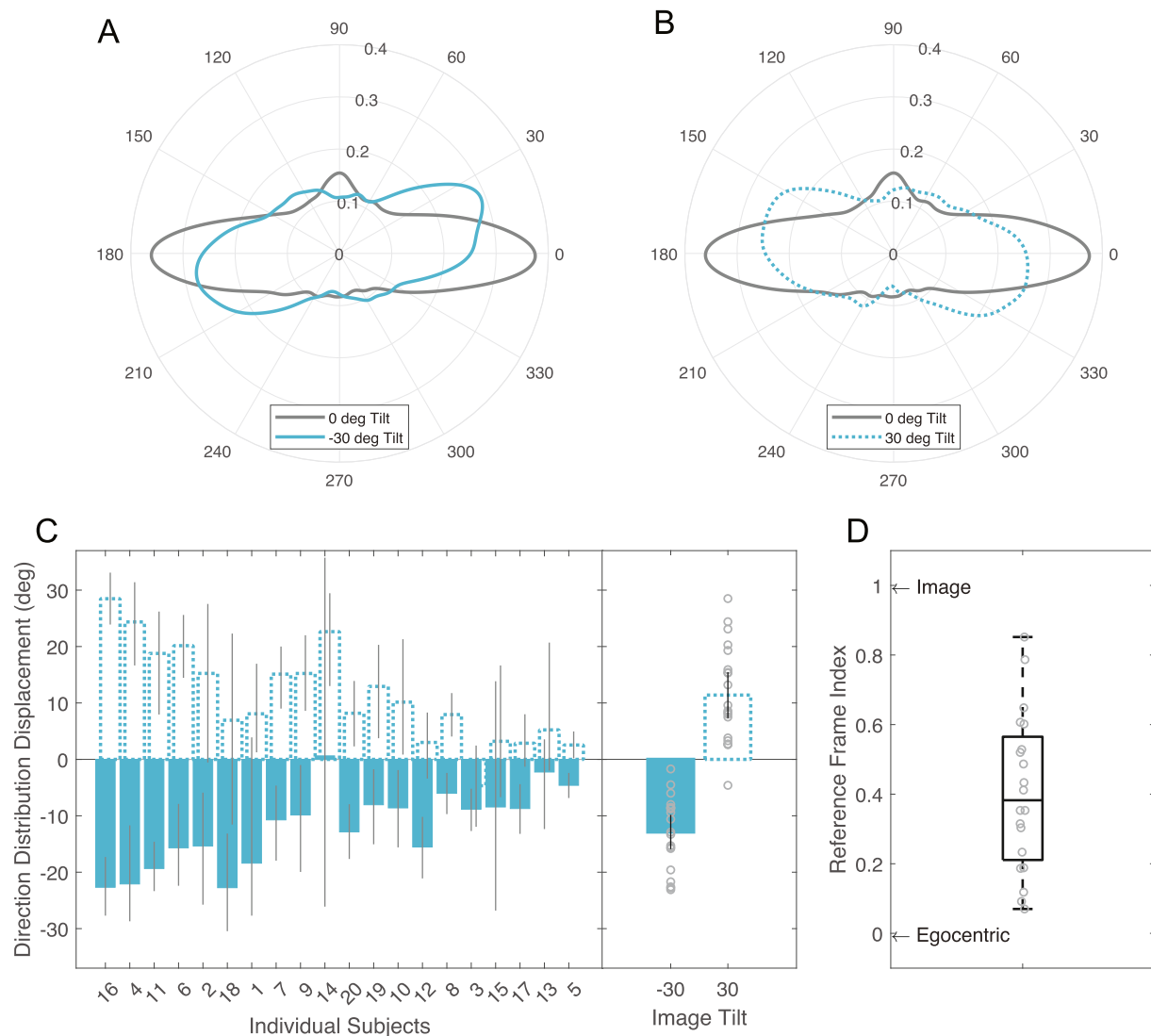


Fig. 2. Saccade polar distributions for all subjects during free viewing for A) -30° and 0° scene tilts, and B) 30° and 0° scene tilts. Radial axis indicates probability and angle indicates saccade direction with 0 angle indicating saccade to the right. C) Left: Average angular displacements for individual subjects with bootstrapped 95% confidence interval error bars. Right: Bar plots showing the direction distribution displacement for all subjects. Circles represent individual data and error bar represents 95% confidence intervals. (D) Box and whisker plot showing the reference frame index derived by scaling the direction distribution displacement values for each subject by the absolute value image tilt in degrees (30°).

world, head, and body and not with respect to the tilted scene.

4. Discussion

Our results show that saccades are more likely to occur in the horizontal direction. However, when a scene is tilted, this preference changes and saccade direction distributions rotate to partially align with the scene. Our study sought to determine whether the effect of scene tilt on saccade directions was amplitude-dependent. We answered this research question by analyzing small and large saccades during free viewing. We found that larger saccades made during free viewing are directed close to parallel to the orientation of a tilted image while smaller saccades are directed with respect to the upright body and head. In this way, large saccades are closely oriented to the tilt of the image while small saccades generally remain horizontal relative to the head. We also observed that fixational saccades were biased horizontally and also unaffected by the presence of a tilted image in the background even though that tilted image did have an effect on ocular torsion.

There are a few reasons why large and small saccades may show differences in direction preference in response to scene tilt. First, large

saccades are useful to bring the fovea to an entirely new region of interest or a new object while small saccades are used to carefully inspect the region or object. Second, large saccades decorrelate neural input more than short saccades by moving receptive fields into areas of the visual scene that are uncorrelated with the previous location [43]. Lastly, large and small saccade targets in visual scenes have different spatial statistics: short saccades are more likely to target regions of scenes with high spatial frequency while long saccades are more likely to target regions of scenes with low spatial frequency [46]. Perhaps, when we use small saccades to inspect a small part of an image, image features showing gravity, cardinal biases in spatial frequency and orientation, and horizontal object relationships are less relevant than when we make larger saccades across the scene.

Previous studies have shown that saccade directions during free viewing are oriented to and aligned with the image. Other studies have sought to determine whether the way the image is framed matters [1]. Here we intended to use the condition with the highest potential to induce a rotation in the fixational saccade direction distributions and that is why we presented the image in a square frame that tilted along with the image. Previous studies used an angle of 45° instead of 30° and

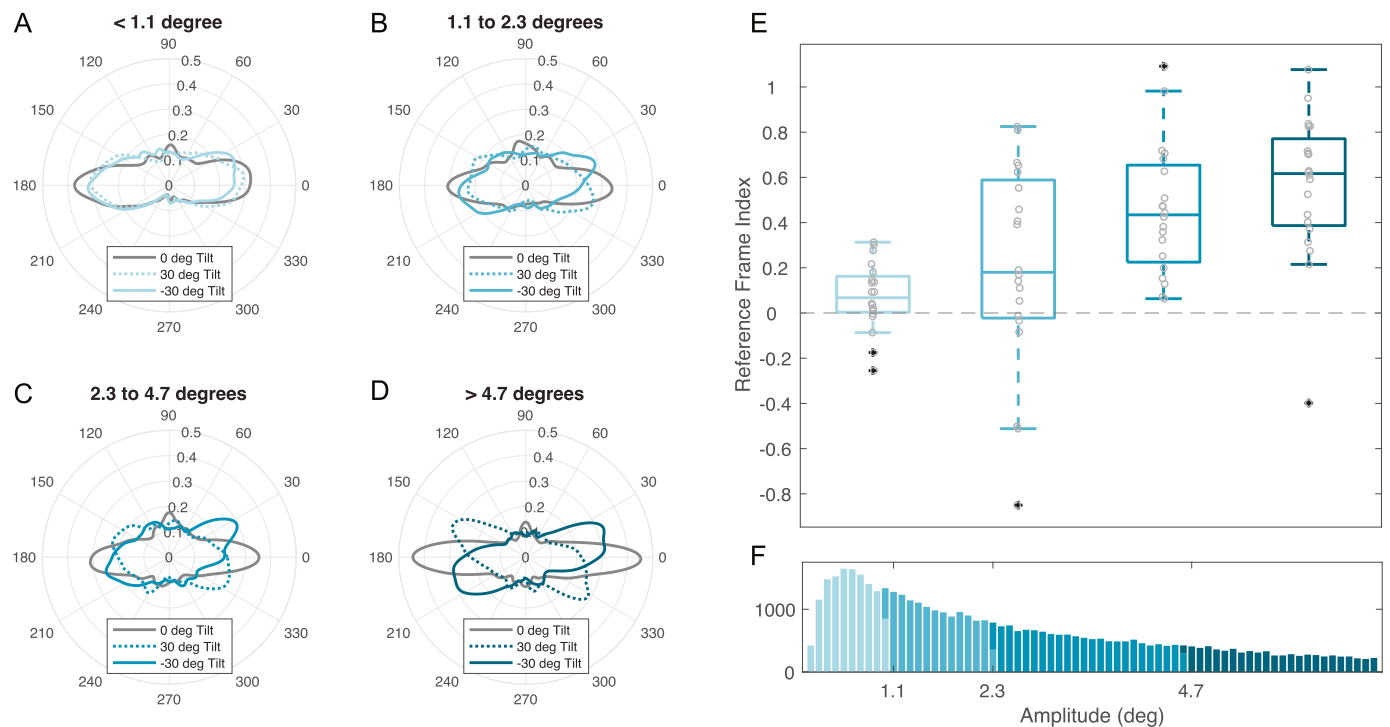


Fig. 3. A-D) Polar distributions of saccade directions during free viewing filtered by amplitude. Amplitudes were determined based on quartiles of the data. Radial axis indicates probability and angle indicates saccade direction with 0 angle indicating saccade to the right. E) Reference frame indices show that larger amplitude saccades were closely oriented to the tilt of the image (closer to 1) while smaller amplitude saccades were oriented egocentrically (closer to 0). F) Amplitude distribution for free viewing saccades, color-coded by amplitude thresholds. Distribution is truncated at 7° for clarity, see Fig. 1 for a full distribution. Vertical axis indicates number of saccades within each bin.

found what appeared to be a much larger rotation of the distributions [1,2,9]. Our study participants were naïve to the experimental manipulation, and when asked about the goal of the experiment, only 35% were able to approximately guess the research question. Additionally, our data was highly variable across subjects: some subjects rotated close to fully with the image tilt while others rotated minimally or partially with image tilt. Future studies should explore the source of such variability.

Saccades share many characteristics across different tasks, such as the velocity amplitude relationship (main sequence) and intersaccadic interval properties [35]. However, saccade characteristics will also vary according to the task subjects are asked to do, be it free viewing, refixating, catching up during smooth pursuit, or eye-head shifts. Here, we studied two different tasks, free viewing and fixation, with different task demands. Fixation requires the observer to intently look at a single stimulus while free viewing requires the observer to simply look at and explore different parts of a scene. These different task demands result in different retinal input, different perceptive experiences, and different sets of data. For example, when looking at amplitude distributions between zero and one degree for each of the two tasks, it is obvious that the resulting distributions are different (e.g., see Figs. 1E and 3F). However, despite the many differences between free viewing and fixation tasks, it is interesting to note that in our data, small saccades less than one degree in amplitude were primarily horizontal in direction and unaffected by the tilt of the scene across both tasks. This result may be coincidental and not necessarily underlie the same mechanisms for the horizontal bias. Future research may benefit from a fuller set of experiments to determine whether the mechanism for this robust horizontal small saccade bias is a general property of small saccades or a byproduct of particular tasks.

There are several ways in which biases in the properties of the visual scenes could be related to the biases we observe in saccades. First, orientation features in images contain certain environmental cues about

the direction of gravity (e.g., floor, walls, horizon). These orientation features can affect both torsional eye position [12] and perception of upright [15] and may interact with saccade generation by aligning the reference frame of saccade generation to the perceived upright orientation. Second, the statistics of image features are not random. Since spatial frequencies and orientations are not equally distributed in all directions in natural scenes, it could be that the visual system has developed a bias in its visual exploration strategy to be optimal with respect to the statistics of the images [41]. Finally, scenes contain objects that are more salient or contain more meaningful information and therefore are fixated more often [18]. If such objects in natural scenes consistently appear at certain locations or are aligned with other objects in recurrent patterns, the saccades made between these objects will present directional biases. All these explanations would predict that if the saccade biases are fully caused by features or statistics in the image, the saccadic distributions would rotate fully along with the image. However, consistent with our data, previous research on the prediction of saccadic movements [27] has found that image properties themselves do not appear to fully explain the biases observed in the data. Instead, those models have had to either explicitly incorporate the saccade bias [29] or learn it from data collected while subjects scan images [47].

It is currently unknown how the horizontal saccade direction bias is implemented in the brain. In the case of other biases, such as the oblique effect where we have a higher sensitivity to precisely report the orientation of a stimulus that is aligned with the cardinal directions, the bias seems correlated with a higher number of neurons that prefer to fire for those cardinal directions [30,44]. In the case of the oculomotor system, the final pathways in the brain stem are segregated along cardinal directions while higher order areas encode possible saccade movements in two-dimensional neuronal maps. While some of these maps appear to encode saccades in a rigid egocentric reference frame (e.g., superior colliculus), others appear to encode saccades in reference frames that are more flexible and may even encode saccades relative to allocentric

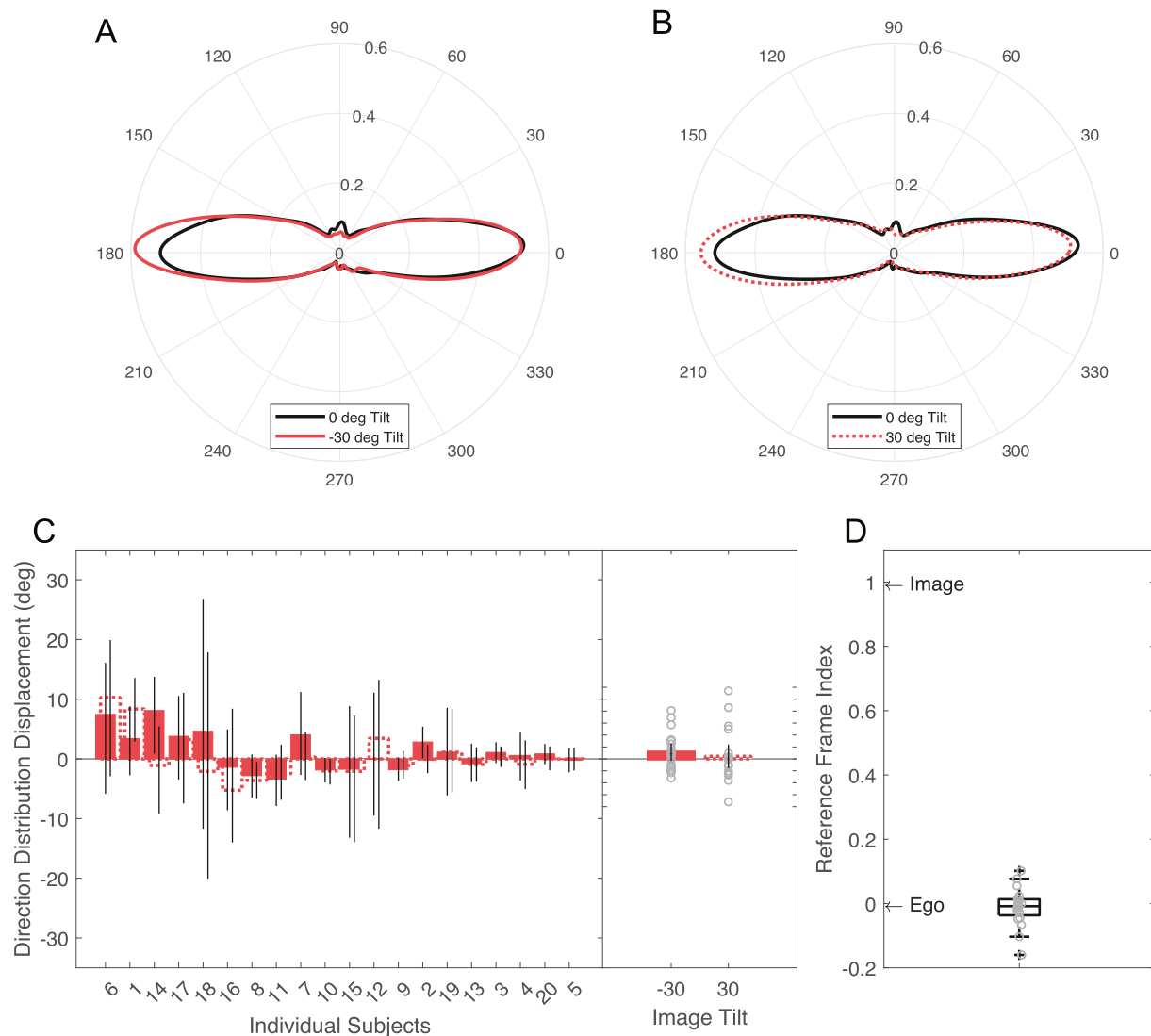


Fig. 4. Fixational saccade polar distributions for all subjects during fixation for A) -30° and 0° scene tilts, and B) 30° and 0° scene tilts. Radial axis indicates probability and angle indicates saccade direction with 0 angle indicating saccade to the right. C) Left: Average angular displacements for individual subjects with bootstrapped 95% confidence interval error bars. Right: Bar plots showing the direction distribution displacement for all subjects. Circles represent individual data and error bar represents 95% confidence intervals. (D) Box and whisker plot showing the reference frame index derived by scaling the direction distribution displacement values for each subject by the absolute value image tilt in degrees (30°).

reference frames (e.g., parietal cortex) [28]. While no study has carefully quantified biases in neural populations in those maps encoding horizontal vs. vertical vs. oblique saccades, our results suggests that there may be two types of biases: one that is present even in the absence of a visual scene that follows an egocentric reference frame, and a second that follows an allocentric reference frame of a natural scene. Future models of saccade generation should be aimed at reproducing these different patterns of bias.

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